Stability and Control

Why the centre of pressure belongs behind the centre of gravity, how swivelling the thrust steers a rocket, and how guidance closes the loop.

Isagi · November 2026 · Theory from NASA Glenn's Beginner's Guide to Rockets; ISRO figures checked 9 November 2026

HOW TO USE THIS CHAPTER

- **Print single-sided** and write in the margins.
- Do each example with a calculator first. Moment = force × distance (use kN·m).
- ISRO does not publish LVM3's centre of gravity, gimbal angles or moments of inertia. Every number in the worked examples is a labelled **what-if** chosen to show the method.

By the end of this chapter you can: explain static stability with CG and CP; compute a stability margin in calibres; compute the side force and turning moment from a gimballed engine; and describe the sensor, computer and actuator loop that steers a launch vehicle.

1 Two points that decide everything

Chapter 0 introduced two points on a rocket. The **centre of gravity (CG)** is the balance point; weight acts through it, and the rocket rotates about it. The **centre of pressure (CP)** is the point where the total air force on the body acts, taken as one. Because propellant burns away, the CG moves during flight (which way depends on where the tanks sit). The CP moves with speed and shape.

NASA's stability discussion gives the key rule. Suppose a gust or a slight tilt turns the nose off the flight path. Air then strikes the side of the rocket, and the sideways air force acts at the CP. If the CP is **behind** the

CG, that force swings the tail away and turns the nose back into the airflow, like the feathers on an arrow. If the CP is **ahead** of the CG, the same force turns the nose further away and the rocket tumbles.



Figure 1. A side force at the CP turns the rocket about the CG. Behind the CG it restores the nose to the flight direction; ahead of it, it makes things worse.

This is why fins are placed at the tail: they add air force far behind the CG and drag the CP aft. Model rockets are often built to keep the CP one to two body diameters behind the CG, a margin measured in **calibres**:

```
stability margin  =  ( distance of CP behind CG ) / ( body diameter )
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> **ISRO LENS: NOT EVERY BIG ROCKET IS STATICALLY STABLE**
> Large launch vehicles often fly with a CP close to or ahead of the CG for part of the flight, because a long body with a heavy payload fairing up front shifts the CP forward. They stay on course not because the air steadies them, but because guidance and an engine that can swivel correct the tilt faster than it grows. Control replaces stability. NASA's guide describes both approaches: fins for a naturally stable rocket, or active steering for one that is not.

2 Steering by swivelling thrust

Chapter 0 said thrust normally acts along the axis through the CG, so it makes no turning effect. Swivel the nozzle by a small angle θ and the thrust gains a sideways component, F × sin θ, acting at the tail, a

distance L from the CG. That produces a turning **moment**:

```
side force  =  F × sin θ
moment      =  F × sin θ × L          (about the CG)
```

This is **thrust vector control**. NASA notes the old German V-2 did it with vanes in the exhaust; modern vehicles swivel the nozzle itself, called **gimballing**. Solid motors cannot swivel a rigid nozzle easily, so large solids use a nozzle on a flexible joint (a flex nozzle). Publicly, ISRO steers the S200 boosters with a flex nozzle and the liquid engines by gimballing, which is consistent with each stage's type (Chapter 0).

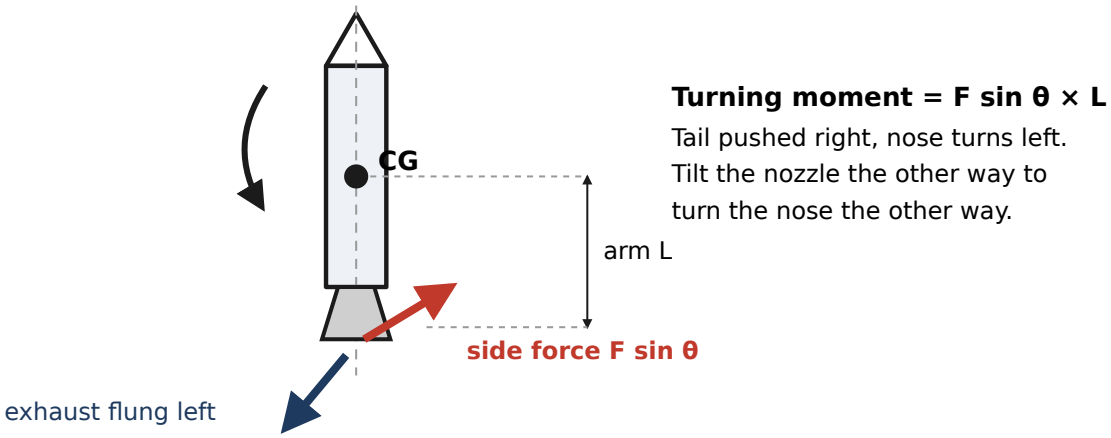


Figure 2. Gimballing. A small nozzle tilt (exaggerated here) gives a sideways force at the tail, and with the arm L to the CG that turns the rocket. Schematic.

Roll (spin about the long axis) is handled by tilting engines differentially or by small separate thrusters. Pitch and yaw come from the same swivelling.

3 Closing the loop: guidance

NASA's fourth system, guidance (Chapter 0), makes this automatic. Inertial sensors (gyroscopes and accelerometers) measure how the vehicle is turning and accelerating. The flight computer compares that with the planned path and commands the actuators. The rocket responds, and the sensors measure again, many times a second.

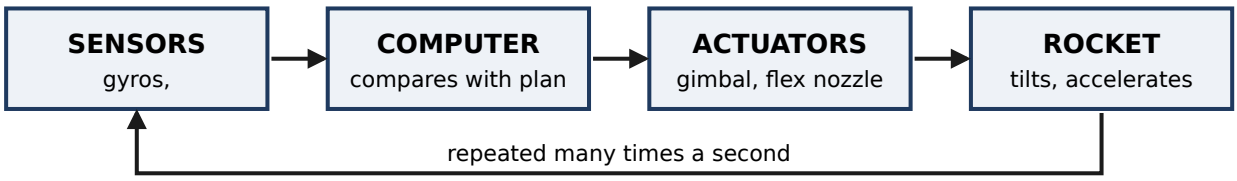


Figure 3. The control loop. In ISRO's description, the equipment bay (EB) at the top of the vehicle houses this guidance equipment.

4 Worked examples (all what-ifs)

ISRO's page gives C25 thrust (200 kN) and the vehicle length (43.5 m). Chapter 1 gave the L110's estimated thrust (about 1.65 MN). Everything else below (CG position, gimbal angle, arm lengths, inertia) is assumed to show the method.

Example A: stability margin in calibres

Take the 4 m body. Assume the CG is 28 m back from the nose and the CP is 31 m back.

```
CP is 31 − 28 = 3 m behind the CG
margin = 3 / 4 = 0.75 calibres
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Positive, so statically stable. Model-rocket practice asks for 1 to 2 calibres, so this vehicle would be marginal by that rule. Real launchers lean on active control.

Example B: C25 gimbal (what-if: 4°, arm 8 m)

```
side force = 200 × sin 4° = 200 × 0.0698 ≈ 13.95 kN
moment     = 13.95 × 8 ≈ 111.6 kN·m
```

About 7% of the 200 kN thrust turns sideways for a 4° tilt. A small angle gives a useful, gentle push.

Example C: L110 gimbal (what-if: 4°, arm 15 m)

```
side force = 1,652 × 0.0698 ≈ 115.2 kN
moment     = 115.2 × 15 ≈ 1,729 kN·m ≈ 1.73 MN·m
```

The L110's moment is about 15 times the C25's. Bigger thrust and a longer arm give far more steering authority in the dense, early part of the climb, when disturbances are largest.

Example D: how fast does the C25 turn? (what-if)

Treat the 44 t of vehicle above the L110 (Chapter 2) as a 20 m uniform rod, moment of inertia $I = mL^2/12$.

```
I = 44,000 × 20² / 12 ≈ 1.47 × 10⁶ kg·m²
angular acceleration α = M / I = 111,600 / 1.47×10⁶ ≈ 0.076
rad/s² ≈ 4.4°/s²
time to turn 10° from rest: t = √(2θ/α) = √(2 × 0.1745 / 0.076) ≈
2.1 s
```

Turning is gentle but fast enough for guidance to correct small errors within seconds. A large vehicle cannot swerve; it nudges.

CHAPTER SUMMARY

- CG is where weight acts; CP is where the air force acts. CP behind CG makes a rocket stable; CP ahead makes it tumble.
- Fins at the tail pull the CP aft. Stability margin in calibres = (CP behind CG) / diameter.
- Many large vehicles rely on active control instead: guidance plus a swivelling thrust.
- Side force = F sin θ; turning moment = F sin θ × L. Solids use flex nozzles; liquid engines gimbal.
- Sensors, flight computer and actuators form a loop repeated many times per second.

Next, Chapter 6: gas physics. Why hot gas expands, why the nozzle works, and what temperature and molecular weight do to exhaust speed.

Practice problems

Answers are on the last page.

P1. A 3 m diameter rocket has its CG 20 m and CP 23 m back from the nose. What is the stability margin in calibres?

P2. An engine of 500 kN gimbals by 3°. What is the side force?

P3. If the arm to the CG in P2 is 10 m, what is the turning moment?

P4. If the same rocket's moment of inertia is 2×10^6 kg·m², what is the angular acceleration from P3's moment? Give rad/s² and °/s².

P5. A gust turns the nose 5° off the flight path of a rocket with its CP ahead of its CG. Describe what happens next without active control.

P6. A solid booster cannot be throttled or shut off, so how can it still help steer the vehicle?

Glossary

Centre of gravity (CG)
The balance point; weight acts through it, and the rocket rotates about it.

Centre of pressure (CP)
The point where the total air force acts.

Static stability
Tendency to return to its heading after a small disturbance.

Calibre
One body diameter; the unit of stability margin.

Thrust vector control
Steering by changing the direction of thrust.

Gimbal
Swivel the engine or nozzle on a mount.

Flex nozzle
A nozzle on a flexible joint that can tilt; used on large solids.

Moment
Force × distance; a turning effect.

Moment of inertia (I)
How hard a body is to spin; $\alpha = M/I$.

Pitch, yaw, roll
Nose up/down, nose left/right, spin about the long axis.

Answer key

Fold this page back before you start the problems.

P1. (23 − 20) / 3 = 1.0 calibre.

P2. 500 × sin 3° = 500 × 0.0523 ≈ 26.2 kN.

P3. 26.2 × 10 ≈ 262 kN·m.

P4. α = 261,700 / 2×10^{6} ≈ 0.131 rad/s^2 ≈ 7.5°/s^2.

P5. Air hits the side and the force acts ahead of the CG, so it turns the nose even further off course. The error grows and the rocket tumbles unless guidance and the engine swivel correct it.

P6. It is steered by tilting the nozzle (a flex nozzle), which redirects the thrust without changing its size.

Sources

Theory: NASA Glenn Research Center, Beginner’s Guide to Rockets (stability, centre of gravity and pressure, thrust vector control, guidance). ISRO: LVM3 launch vehicle page (C25 thrust, vehicle length, equipment bay). The use of flex nozzles on the S200 and gimballed liquid engines is from public ISRO descriptions. The CG, CP, gimbal angles, arm lengths and inertia in the examples are assumptions for illustration; ISRO does not publish them.